Petr **Šimčák**

Profile

Software developer with strong foundations in C, C++, and Python through 42 Prague and academic project work. Experience includes parsing, networking, Unix process handling, Docker-based environments, and signal-processing workflows.

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Technical stack

- C, C++, Python, Bash, MATLAB
- Git, Docker, Linux, Nginx, LaTeX
- Sockets, parsing, Unix processes
- Backend workflows, signal processing

Human languages

Czech (native)
English (fluent)
German (B1)

Links

Cub3D repository
ft_irc repository
Thesis repository

Technical projects

2026–ongoing
FT_TRANSCENDENCE *(full-stack web application)*
◇ Current team-based 42 project focused on a larger end-to-end application.

2026
FT_IRC *(C++98, non-blocking sockets, poll)*
◇ Built a minimal multi-client IRC server in a team of three; responsible for cleaning server-side code and implementing IRC commands.

2025
CUB3D *(C, MLX42, raycasting)*
◇ Implemented map parsing and core raycasting logic in a simplified Wolfenstein-style engine.

2025
HEART RATE ESTIMATION FROM PPG SIGNALS *(Python, signal processing)*
◇ Compared multiple HR-estimation methods, including a Hjorth descriptor-based approach, and worked on automatic PPG signal-quality classification.

2024–2025
ADDITIONAL 42 PROJECTS *(C, Docker, Linux)*
◇ minishell: Unix processes, pipes, redirections. Inception: containerized multi-service setup with Docker and Nginx.

Education

2023–2026
42 PRAGUE *(42 Heilbronn: 2024–2025)*
◇ Project-based software engineering curriculum with emphasis on low-level programming, problem solving, and peer learning.

2020–2025
BRNO UNIVERSITY OF TECHNOLOGY
◇ Bachelor's Degree in Sport Technology. Thesis: *Heart Rate Estimation from the PPG Signals*.

Additional experience

2026–present
TUTOR AT 42 PRAGUE
◇ Support students with debugging, project understanding, and technical problem solving.

2025
ŠKODA B3R BRIGADE — *Team Lead / Backend Developer*
◇ 2nd place at Škoda Hackathon; backend work on a talent-matching application using FastAPI, React, Docker, and SQLite.

2022–present
3D PRINTER MAINTENANCE AND SUPPORT
◇ Assembled a 3D printer at BUT and later maintained and supported it at 42 Prague.

2013–2021
FILM / THEATRE BACKGROUND
◇ Earlier professional experience in collaborative performance environments; developed communication, discipline, and adaptability.